1.3. Summenzeichen

(a) $\sum_{k=0}^{2} k^2 = 0 + 1 + 4 = 5$

(b) $\sum_{k=-1}^{3} 2^{k-1} = 1/4 + 1/2 + 1 + 2 + 4 = 1/4 + 2/4 + 4/4 + 8/4 + 16/4$

$= \frac{1 + 2 + 4 + 8 + 16}{4} = 31/4$

(Bemerkung: $\sum_{k=-1}^{3} 2^{k-1} = \sum_{u=0}^{4} 2^{u-2}$ "Index-Shift")

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$1/4 + 1/2 + 1 + 2 + 4$

(c) $\sum_{k=0}^{4} (1/2)^k = (1/2)^0 + (1/2)^1 + (1/2)^2 + (1/2)^3 + (1/2)^4$

$= 1 + 1/2 + 1/4 + 1/8 + 1/16$

$= \frac{16 + 8 + 4 + 2 + 1}{16} = 31/16$

(d) $\prod_{k=0}^{10} 2^k = 2^0 \cdot 2^1 \cdot 2^2 \cdot 2^3 \cdot 2^4 \cdot 2^5 \cdot 2^6 \cdot 2^7 \cdot 2^8 \cdot 2^9 \cdot 2^{10}$

$= 2^{0+1+2+3+4+5+6+7+8+9+10} = 2^{55}$